

Trust the Card, Not the Resemblance

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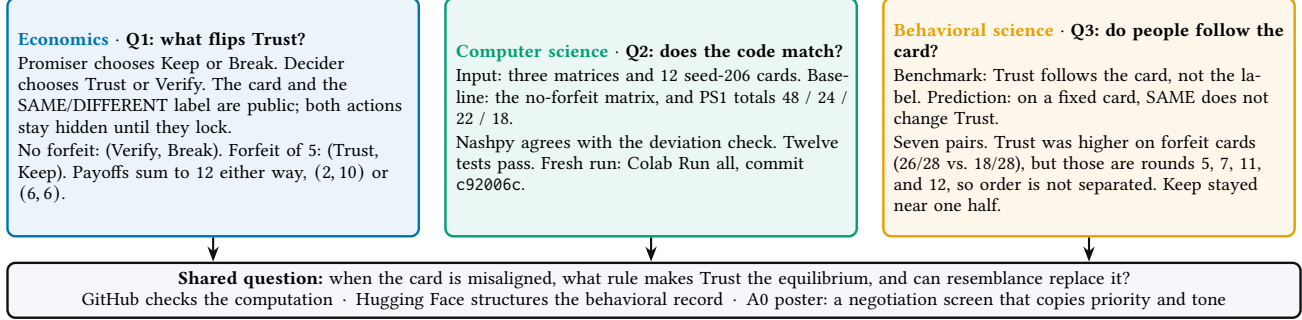


Figure 1: Three linked panels share one question, the forfeit rule. Completed model evidence is the matrix equilibria, the Nashpy check, and the PS1 replay of $p^* = 0.6$. Human play is seven pairs: the Trust rise is descriptive and mixed with order; Keep stayed near one half. Interface clicks are not those sessions. Panel colors, also named in the text: Strategy #0072B2, Interdisciplinary #009E73, Leadership #E69F00; arrows and the lower bar use Ink #111111.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** FP12. **Symposium:** A.

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

1 Question and verified gap

An AI negotiation assistant may copy a user’s priorities and tone. The shared question is: when the payoff card is misaligned, what rule makes Trust an equilibrium even when a public SAME/DIFFERENT label is present? Q1 asks what the Nash equilibrium is before and after a 5 forfeit. Q2 asks whether Nashpy returns the same equilibria and whether the PS1 benchmark and seed 206 can be reproduced. Q3 asks whether two real players follow the card or the similarity label. Seven pairs have played.

PS1 supplied the threshold $p^* = 0.6$ [2], but its Promiser did not choose an action. Cheap talk does not change the Break cell [3]. In the trust game the second mover chooses a return [1]; here the card makes one action strictly better before play. A general penalty is also incomplete unless the rule says that a similarity cue cannot cancel it. This paper turns both sides into players and tests one narrow intervention.

Figure 1 puts Q1’s equilibrium, Q2’s reproduction, and Q3’s seven pairs on one evidence chain. The shared object is the forfeit rule. The value $p^* = 0.6$ is the PS1 benchmark reproduced in the computer-science panel. It is not an input to the new Nash equilibrium.

2 Strategic model and three lenses

The Promiser privately chooses Keep or Break, and the Decider chooses Trust or Verify. Both first lock their choices; before the reveal, neither can see the other’s action, so the stage is simultaneous.

The public score card is either aligned or misaligned: on the aligned card, Keep/Break payoffs are 6/3; on the misaligned card, they are 6/10 before the forfeit. Separately, SAME/DIFFERENT is a visible label showing whether the Promiser’s displayed priority matches the Decider’s priority. It changes no payoff and cannot waive a forfeit. Computer code deals the cards, hides choices, scores the round, and records the result; it does not choose for either person. The game is repeated for twelve rounds with a fresh card each time; scores do not enter the next card. Players can remember earlier rounds, so the repetition is not a reputation model with carried payoffs.

The Decider’s payoffs are 6 after Trust and Keep, −4 after Trust and Break, and 2 after Verify. The Promiser’s Keep/Break payoffs are the same in Trust and Verify rows: on an aligned card they are 6/3; on a misaligned card they are 6/10 before the forfeit and 6/5 after it. Thus the Promiser’s payoff does not depend on whether the Decider trusted. SAME/DIFFERENT changes no number.

	Keep	Break		Keep	Break
Trust	(6, 6)	(−4, 3)	Trust	(6, 6)	(−4, 10)
Verify	(2, 6)	(2, 3)	Verify	(2, 6)	(2, 10)

Aligned card (left) and misaligned card without a forfeit (right); each cell is (Decider, Promiser).

	Keep	Break
Trust	(6, 6)	(−4, 5)
Verify	(2, 6)	(2, 5)

Misaligned card with a forfeit of 5. The equilibrium is (Trust, Keep).

On the aligned card, Keep strictly dominates Break and the unique Nash equilibrium is (Trust, Keep), with (6, 6). On the misaligned card without a forfeit, Break strictly dominates Keep; against

Break the Decider prefers Verify because $2 > -4$. The unique equilibrium is (Verify, Break), with (2, 10). If the Promiser deviates to Keep, its payoff falls from 10 to 6; if the Decider deviates to Trust, its payoff falls from 2 to -4.

The intervention is a 5-point forfeit only on the misaligned card when Break occurs. It is larger than $10 - 6 = 4$, so Keep strictly dominates Break and the unique equilibrium becomes (Trust, Keep), (6, 6). The Promiser's deviation falls from 6 to 5 and the Decider's deviation from 6 to 2. The two equilibria have the same total payoff, 12: (2, 10) before the intervention and (6, 6) after it. The 5 points are lost by the Promiser, not transferred to the Decider; on the Trust-and-Keep path they are not paid, so the rule changes distribution rather than the two players' total. If the Break payoff exceeds 11, or collection fails, the flip no longer follows. Because the card is public and Keep or Break is selected by strict dominance, the new equilibrium does not need a probability input. The $p^* = 0.6$ value is PS1's benchmark: it answers when Trust and Verify would be equally good if keeping were still a probability. It is not an input to this new equilibrium.

The game-theory lens identifies strict dominance and the unilateral deviations; the equilibrium language follows Nash [6] and the normal-form treatment in Osborne and Rubinstein [7]. The social-choice lens makes the trade-off visible: the Decider gains 4 while the Promiser loses 4 when the misaligned equilibrium changes; no rule here makes both strictly better. The mechanism is small: forfeit 5 only on a misaligned Break, and SAME cannot waive it. The Promiser's payoff does not depend on Trust, so one action strictly dominates. It is still a two-player game: the Decider's best response uses that action, and the forfeit is what changes it. PS1 had no Promiser strategy. A copied priority on a ballot raises the same label problem; this paper does not add a voting game.

3 Computation and auction boundary

Nashpy returns the three equilibria in Section 2, and the deviation checks agree. Decider rows are (6, -4), (2, 2) in each matrix. The PS1 seed-206 replay, not a human sample, totals 48, 24, 22, and 18. Twelve tests pass, including JavaScript-Python agreement. Equilibrium play of the new 12 cards, with no people, scores 56 and 88. That schedule balances SAME and DIFFERENT, so the model's Trust gap is 0. In Section 4, SAME Trust (28/42) is below DIFFERENT Trust (32/42), opposite the direction in which a familiar label would raise Trust.

Links: <https://github.com/dku-comsci-econ206-Autumn2026/PS2-Muhan> (commit c92006c), https://colab.research.google.com/github/dku-comsci-econ206-Autumn2026/PS2-Muhan/blob/main/PS2_trust_forfeit.ipynb, https://huggingface.co/spaces/dku-comsci-econ206-2026/Trust_Forfeit. Colab: Runtime → Run all. Bids (9, 6) and (8, 3) give second bids 6 and 3: Trust, Keep, then Verify, Break. This auction note is not the contribution. Poster file: PS2-FP12-TrustForfeit-A0.

4 Behavioral comparison

The model predicts Trust rate 0 and Break on a misaligned card with no forfeit, and Trust rate 1 and Keep when the card is aligned or the forfeit is 5. SAME should not change Trust [4, 5].

Seven pairs, session code 206, one computer each, and the page hides the Promiser's choice until the Decider locks. All 84 rounds have both choices (data/session-01.csv-session-07.csv). Interface clicks are separate. Intervals treat rounds as independent and are too narrow; Trust by pair runs from 6/12 to 11/12.

Card	Prediction	Observed
Aligned	Trust 1, Keep 1	Trust 16/28 (0.39–0.74); Keep 14/28 (0.33–0.67)
Misaligned, no forfeit	Trust 0, Break 1	Trust 18/28 (0.46–0.79); Break 13/28 (0.30–0.64)
Misaligned, forfeit 5	Trust 1, Keep 1	Trust 26/28 (0.77–0.98); Keep 15/28 (0.36–0.71)

Trust did not separate an aligned card from a misaligned card with no forfeit: 16/28 against 18/28, and those intervals overlap. The rise from 18/28 to 26/28 when the forfeit appeared is descriptive only, and those cards are rounds 5, 7, 11, and 12, so it is mixed with order. Keep was 14/28, 15/28, and 15/28, about half on every card. On an aligned card Break pays 3 against 6, so that Break has no strategic reason. On forfeit cards Keep is 15/28 ≈ 0.54 , below $p^* = 0.6$. Trust then expects $10 \times 15/28 - 4 \approx 1.36$, below Verify's 2, yet the Decider trusted in 26/28 of those rounds. Keep overall is 44/84, about 0.52, and below 0.6 on every card, so after the fact Verify was better on each card. Always Verify scores 24 in a session; these Deciders averaged 19.7. That 24 is also the PS1 always-Verify total. They did not know the Keep rate when they chose. SAME Trust was 28/42, below DIFFERENT at 32/42. Across pairs the forfeit raised Keep for 2, left 2 unchanged, and lowered it for 3. One person is the Promiser for all 12 rounds. No written consent is recorded; this was a course exercise, and the files contain no names. A familiar label can stand in for keeping the promise [4, 5]; these pairs did not show that. The matrix remains the benchmark.

5 Application, impact, and open science

The case is a procurement screen that copies the buyer's priority and tone. In the matrix the forfeit makes Break unattractive; in the seven pairs it did not raise Keep. Show the card first, and do not let similarity waive the forfeit.

Open Science Statement

<https://github.com/dku-comsci-econ206-Autumn2026/PS2-Muhan> (commit c92006c); https://colab.research.google.com/github/dku-comsci-econ206-Autumn2026/PS2-Muhan/blob/main/PS2_trust_forfeit.ipynb; https://huggingface.co/spaces/dku-comsci-econ206-2026/Trust_Forfeit. The seven sessions are in data/. Interface clicks are separate. Runtime → Run all reproduces the matrices. The 2056 note is in the horizon appendix.

Statement of Contribution to the UN's SDGs

SDG 10 asks who could pay a real-money forfeit of 5. SDG 16 asks whether disclosure and appeal make the rule contestable. Neither impact is claimed.

Submission milestones

Review-ready: 27 September 2026, 11:00 p.m. Symposium and peer review: 28 September 2026, 2:45–5:15 p.m., IB 2050. Final revision: 30 September 2026, 11:00 p.m. All times are Beijing time.

A Author notes

Acknowledgements. I thank Professor Luyao Zhang. The PS1 final review asked for a two-sided game and visible matrices; this final revision implements both. On 2026-09-29 I withdrew the chatbot-subsidy-auction draft because it did not continue the PS1. The 28 September discussion was about a withdrawn draft of a different model. Those comments are not reviews of this paper, so they are not quoted here.

Contribution. Muhan Chen is the sole author. My intellectual decisions were to make the Promiser a real player, set the forfeit at 5 because the gain from Break is 4, keep similarity outside the matrix, and make two humans play the game. Outputs are the GitHub, Colab, and Hugging Face artifacts. Verification is the dominance proof, the three Nashpy checks, and the seed-206 replay. The human sample is seven pairs in the repository folder `data/`.

References

- [1] Joyce Berg, John Dickhaut, and Kevin McCabe. 1995. Trust, Reciprocity, and Social History. *Games and Economic Behavior* 10, 1 (1995), 122–142.
- [2] Muhan Chen. 2026. Pricing a Promise: When Similarity, Not Incentives, Buys Trust in AI Negotiation. COMSCI/ECON 206 PS1 research proposal, Duke Kunshan University.
- [3] Vincent P. Crawford and Joel Sobel. 1982. Strategic Information Transmission. *Econometrica* 50, 6 (1982), 1431–1451. doi:10.2307/1913390
- [4] Lisa M. DeBruine. 2002. Facial Resemblance Enhances Trust. *Proceedings of the Royal Society B: Biological Sciences* 269, 1498 (2002), 1307–1312. doi:10.1098/rspb.2002.2034
- [5] Daniel Kahneman. 2007. Maps of Bounded Rationality: A Perspective on Intuitive Judgment and Choice. In *Nobel Lectures: Economic Sciences 2001–2005*. World Scientific. Nobel lecture delivered 2002.
- [6] John F. Nash. 1950. Equilibrium Points in n -Person Games. *Proceedings of the National Academy of Sciences* 36, 1 (1950), 48–49. doi:10.1073/pnas.36.1.48
- [7] Martin J. Osborne and Ariel Rubinstein. 1994. *A Course in Game Theory*. MIT Press, Cambridge, MA.

B Parameters and reproducibility

Players. Two humans: Promiser (Keep/Break) and Decider (Trust/Verify).

Score cards. Aligned: Promiser Keep/Break payoffs 6/3; misaligned: 6/10 before the forfeit. **Labels.** SAME if the displayed priorities match; DIFFERENT otherwise. The label changes no payoff and cannot waive the forfeit. **Decider payoffs.** Trust/Keep 6, Trust/Break −4, Verify 2. **Promiser payoffs.** Aligned: Keep 6, Break 3; misaligned: Keep 6, Break 10, or 5 after the forfeit. **Information.** Public card and label; actions hidden until both lock; simultaneous action. **Horizon.** Twelve fresh rounds; no carry-over score. **Benchmark.** $10p - 4 = 2$, so $p^* = 0.6$. **Computation.** Nashpy for matrices; Python/Colab for replay; interface test marked simulated.

Keep is strictly better than Break on the aligned card ($6 > 3$). On the misaligned card without a forfeit, Break is strictly better ($10 > 6$), so the Decider’s best response is Verify ($2 > -4$). With the 5 forfeit, Keep is strictly better ($6 > 5$), and Trust is then the Decider’s best response ($6 > 2$). Nashpy returns exactly the three pure equilibria named in Section 3. Reproduction uses commit `c92006c`, the matrix test, and Colab Runtime → Run all; the saved seed-206 output matches 48/24/22/18. The full hash is in the links section.

AI-use disclosure. 2026-09-29 to 2026-09-30, Cursor coding agent. Purpose: repository, notebook, Hugging Face game, and revision prompts from the author’s locked model. Accepted: human Promiser and Decider, forfeit of 5, the three matrices, and Nashpy.

Rejected: an AI player, a personal Drive Colab citation, and invented human data.

C Pseudocode and horizon record

Pseudocode. (1) Read the score card and whether the forfeit rule is active. (2) If the card is aligned, set the forfeit to false; SAME/DIFFERENT never changes it. (3) Fill the 2×2 payoff matrices from the stated Decider and Promiser payoffs. (4) Use strict dominance and best responses to mark the unique pure Nash equilibrium, or record “not unique” at a 4-point second bid. (5) Replay seed 206 and record PS1 totals 48, 24, 22, and 18. (6) For the appendix comparison, record only the second bid and return the > 4 , < 4 , or $= 4$ outcome.

PS1. $p^* = 0.6$; Keep was a draw. **PS2.** Forfeit 5 flips the matrix. In seven pairs, Trust was higher on forfeit cards, descriptively and mixed with order; Keep did not move. **Next.** Do not treat interface clicks as behavior. **2056.** A checked rule for when a copied style cannot replace the payoff card.

The 2056 hope is to be remembered for a testable rule: in public or commercial negotiation, similar style cannot replace the payoff card’s forfeit. It builds on the PS1 threshold, these three equilibria, and seven exploratory pairs, not on a claim of a Nobel prize or completed SDG impact.

D Four-source literature record

Chen, PS1, 2026. $p^* = 0.6$; similarity changes no payoff; Keep is a draw. This paper lets the Promiser choose, and a forfeit of 5 flips the misaligned equilibrium. **Crawford and Sobel 1982.** Cheap talk need not change the Break payoff. The forfeit changes that cell, and the label cannot waive it. **Berg, Dickhaut, and McCabe 1995 [1].** The second mover chooses a return. Here the card makes Keep or Break strictly dominant before play. **Nash 1950; Osborne and Rubinstein 1994.** The equilibrium definition is applied to these three matrices and checked in Nashpy.

E PS1 baseline and revision

PS1’s threshold was $p^* = 0.6$ and its similarity cue did not alter payoffs. Its Keep probabilities were exogenous. The final review dated 2026-09-25 reports 75.57/100, Research Quality $Q = 59/100$, Research 50.57/60, and Revision 0/15 because an implemented v1-to-v2 change was not visible. The seed-206 totals copied from the graded PS1 are benchmark +48, always-Verify +24, always-Trust +22, and Trust-if-similar +18. The present revision makes Promiser action endogenous and adds the 5 misaligned-card forfeit; the result changes from a decision threshold to a Nash prediction.

F Pair-level human record

Each row is one file in `data/`. Trust and Keep are counts out of 12 rounds. The last column is Keep on the four forfeit cards minus Keep on the four misaligned cards with no forfeit.

Pair 1: Trust 9/12, Keep 6/12, forfeit minus no forfeit −1. Pair 2: 7/12, 8/12, −1. Pair 3: 6/12, 7/12, 0. Pair 4: 9/12, 8/12, +2. Pair 5: 8/12, 5/12, 0. Pair 6: 10/12, 4/12, +1. Pair 7: 11/12, 6/12, −1.

G Three-lens stress test

Game theory. If many Promisers Keep without the forfeit, strict dominance does not describe them. That check is triggered: on

misaligned cards with no forfeit they Keep in 15/28 rounds, and with the forfeit of 5 they still Break in 13/28. **Social choice.** If the Decider’s session score does not reach the equilibrium total of 56, that trade-off is not what these pairs received. Their mean session score is 19.7. **Mechanism design.** The label cannot waive the forfeit in the rules. They Kept in only 15/28 rounds when the forfeit was on the card. The rule was displayed; the Promiser’s behavior did not implement it.

H Revision map from the final PS1 review

Matrix. Three matrices are printed in Section 2, including the forfeit-of-5 card. The README prints the same three. Commit c92006c. **Both sides act.** The Promiser now chooses Keep or Break. Seven pairs are in Section 4. **Information.** The card is public and the actions are hidden until both lock. Scores do not carry over. **Nash.** Strict dominance, best responses, and deviations are in Sections 2–3. Only pure strategies are claimed. **Nashpy.** The output matches all three matrices. **Separate evidence.** Section 4 separates the equilibrium replay, interface clicks, and seven human pairs. The 84 rounds are valid; the intervals are descriptive. **One commit.** The course repository, Colab notebook, and Hugging Face Space use the hash in the links section. **Keep.** The two human roles and the matrices stay. The seven pairs do not replace the matrix. **Revise.** Figure 1 is the evidence chain. Trust was higher on forfeit cards, descriptively and mixed with order; Keep did not move. **Investigate.** An Innovation Studio run was not completed, so no Studio URL is cited. The benchmark, forfeit cell, overturning condition, and illustrative-score limit are in the evidence section.

I Evidence and withdrawn draft

The benchmark is $p^* = 0.6$, and Trust follows only when Keep is strictly better. Cheap talk has no forfeit cell, and PS1’s Promiser had no choice. The flip fails if Break’s gross payoff is greater than 11 or the forfeit cannot be collected. The payoff numbers in the matrices are illustrative. The human scores are observed. The sample is seven pairs. The withdrawn chatbot-subsidy-auction draft is not the model of this paper.

J Links and open-science checklist

The final release uses the same organization commit in <https://github.com/dku-comsci-econ206-Autumn2026/PS2-Muhan> (commit c92006c), https://colab.research.google.com/github/dku-comsci-econ206-Autumn2026/PS2-Muhan/blob/main/PS2_trust_forfeit.ipynb, and https://huggingface.co/spaces/dku-comsci-econ206-2026/Trust_Forfeit. The full commit is c92006c9fffc2dd8ab33dd495b25d8f704d40e71. The repository contains the matching matrices, code, parameters, tests, outputs, and MIT license. The notebook exposes the seed, runs the matrix checks, and reproduces +48/+24/+22/+18. The Space is the two-player interface. The seven session files are the human sample. Other clicks are not that sample.

K Eight-row structure summary

Background. This paper: two humans choose Keep/Break and Trust/Verify around an aligned or misaligned card plus a separate similarity label. PS1: Trust/Verify around a public payoff table and

a similarity cue. **Motivation.** This paper: a style-matching negotiation screen can invite trust when the card is misaligned; a forfeit tests a correction. PS1: trust may follow resemblance rather than incentives. **Questions.** This paper: the two misaligned equilibria, whether computation reproduces them and PS1, and whether people follow the card or SAME. PS1: when the trust threshold applies, and whether similarity changes behavior. **Method.** This paper: simultaneous hidden actions, public payoffs, twelve fresh rounds, three matrices, Nashpy, and seven sessions. PS1: exogenous Keep probabilities 0.85/0.25 and a seed-206 replay. **Results.** Actual: three Nash equilibria; PS1 totals +48/+24/+22/+18; seven pairs, Decider match 52/84, Promiser match 42/84. PS1 had no human data. **Merits.** A 5 loss changes strict dominance and leaves similarity outside the table. PS1 separates incentive-based trust from similarity-based over-trust. **Impacts.** Disclosure and a non-waivable forfeit are questions, including SDG 10 and 16. PS1 impacts were not measured. **Revision.** PS1 feedback led to a two-sided human game, visible matrices, Nashpy parity, corrected links, and a split between model output and human play. The review had identified missing strategic actions, a missing matrix, an unclear information structure, and missing model-aligned computation.

L Auction record

This is a bounded allocation record in the appendix, not the paper’s contribution. Two prospective Promisers each submit a nonnegative integer bid for how much they are willing to have taken if they Break on a misaligned card. The two bids compete only to determine who becomes Promiser. The Decider is not selected by the bid. The bid is a private willingness signal, while the score card and SAME/DIFFERENT label are public. SAME/DIFFERENT cannot change a bid or break a tie; a session seed breaks ties.

The higher bidder’s bid determines the role, but the forfeit written into a misaligned Break cell is the second-highest bid. It is paid only after Break, not after Keep. If the second bid is greater than 4, the Break payoff is 10 minus that bid and is strictly below Keep’s 6, so the unique equilibrium is Trust, Keep. If the second bid is less than 4, Break is strictly better than Keep and the unique equilibrium is Verify, Break. If the second bid equals 4, both Keep and Break give the Promiser 6, so there is no unique equilibrium. These cases follow the code’s rule exactly; no claim is made that bidding cost is dominant or that this is a standard independent-private-values auction.

Values or signals. Each bid is a nonnegative integer. The card and the SAME/DIFFERENT label are public. **Prediction.** Second bid > 4: Trust, Keep; < 4: Verify, Break; = 4: no unique equilibrium. **Bid and allocation.** The higher bidder becomes Promiser. The bid does not choose the Decider. A seed breaks ties. **Payment.** On misaligned Break, the Promiser pays the second-highest bid. On Keep, payment is zero. **Stopping.** One sealed pair of bids, then one hidden Keep/Break and Trust/Verify choice. **Utility.** Decider gets 6, −4, or 2. Promiser gets 6 on Keep and 10 minus the forfeit on Break. **Revenue.** If the second bid is greater than 4, equilibrium is Keep and nothing is collected. If it is less than 4, equilibrium is Break and revenue equals the second bid. **Efficiency.** Role assignment ignores similarity. Fairness is unresolved. **Behavior.** This auction

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was not played. The card game has seven pairs. If a later bid sample follows SAME, revise the protocol and keep the matrix.